



Prealgebra includes a thorough exploration of the fundamentals of arithmetic, including fractions, exponents, and decimals. We also introduce beginning topics in number theory and algebra, including common divisors and multiples, primes and prime factorizations, basic equations and inequalities, and ratios. The course also surveys a wide range of topics, including geometry, counting, statistics, and probability.

This course is specifically designed for high-performing students and draws material from many programs for top middle school students in the country. Our philosophy is that students develop more by learning to solve problems they haven't seen before, as opposed to offering repeated drills that students can memorize their way through. In this way, our classes are structured much more like courses at top-tier colleges.

Book: *Prealgebra* by Richard Rusczyk, David Patrick, and Ravi Boppana

Time Commitment: This 36-week course includes 105 minutes in-class and 1-3 hours of homework each week, corresponding to a full year course.

Content:

Week	Topic
1	Creative Calculations I
2	Creative Calculations II
3	Introduction to Exponent Laws
4	Nonpositive Integer Exponents
5	Multiples and Divisibility Tests
6	Primes and Prime Factorization
7	LCM, GCD
8	Fractions I
9	Fractions II
10	Linear Equations
11	Word Problems
12	Midterm 1
13	Inequalities
14	Arithmetic with Decimals
15	Decimals and Fractions
16	Ratios and Proportions
17	Conversions, Speed, Rates
18	Percents

Week	Topic
19	Percent Increase and Decrease
20	Squares and Square Roots
21	Arithmetic with Square Roots
22	Angles and Parallel Lines
23	Angles in Polygons
24	Midterm 2
25	Perimeter and Area
26	More Triangles and Circles
27	Pythagorean Theorem
28	Special Triangles
29	Quadrilaterals
30	Basic Statistics
31	Graphs and Charts
32	Introduction to Counting
33	Counting and Probability
34	Problem-Solving Strategies I
35	Problem-Solving Strategies II
36	Final Exam

Common Core State Standards (Middle School):

Grade	Domain	Standards
6	Ratio & Proportional Relationships	1, 2, 3bcd
	The Number System	1, 4, 5, 6a, 7a
	Expressions & Equations	1, 2abc, 3, 4, 6, 7, 8, 9
	Geometry	1
	Statistics & Probability	4
7	Ratio & Proportional Relationships	1, 2c
	The Number System	1cd, 2abcd, 3
	Geometry	4, 5
	Statistics & Probability	5, 7a, 8ab
	Expressions & Equations	1, 2, 3, 4ab
8	Expressions & Equations	1, 2, 7b
	The Number System	1, 2
	Geometry	5, 6, 7

Common Core State Standards (High School):

Domain	Subdomain	Standards
Number and Quantity	Quantity	1, 2, 3
Algebra	Seeing Structure in Expressions	1, 2, 3
	Creating Equations	1, 4
	Reasoning with Equations and Inequalities	1, 3
Statistics & Probability	Interpreting Categorical & Quantitative Data	1, 2, 3

Sample Problems:

- ▶ Jack finds the product of three different prime numbers. Is it possible for the sum of the digits of Jack's product to be 18? Why or why not?
- ▶ Five workers together can build a road in 20 days. Suppose every worker works at the same rate. If three workers work on the road for 10 days before eleven more workers join them, then how long total will it take to build the road?
- ▶ Rebecca walks 100 feet in a straight line. She then turns 20 degrees to the left and walks another 100 feet, and then turns 20 degrees to the left again. She continues this pattern until she reaches the point where she started. How far did she walk?
- ▶ The mean, median, and mode of the five numbers 5, 7, 8, A , B are equal. (The list has a single mode.) Find all possible values of $A + B$.
- ▶ In a sports league, there are 20 total teams, divided into 4 divisions of 5 teams each. Over the course of a season, each team plays each of the other teams in its own division 3 times, and each of the other teams in the other divisions twice. How many games does the league have in a complete season?